

ASIT KUMAR DANDAPAT



- 1. Contact Information:** Address: AT Kochilapada, PO Madhapur, Dist-Mayurbhanj, Odisha, PIN-757103 Email: asit80t@gmail.com | Mobile: +91 8658003717 Nationality: Indian | Category: OBC (General) | Date of Birth: 14th Jan 1991 | Marital Status: Unmarried

2. Research interests:

- a) Watershed Hydrology & Rainfall-Runoff Modelling – Simulation and analysis of watershed hydrological processes using RS-GIS, R Studio, GMS-WMS.
- b) Drought & Flood Susceptibility Assessment – Including Energy-Food-Water (EFW) nexus analysis (MCDA).
- c) Soil Erosion & Sediment Transport Modelling – Using RUSLE and geospatial techniques.
- d) Groundwater Quality Assessment – Applying EWQI, Entropy Weight Method, and TOPSIS.
- e) Land Use / Land Cover Analysis – Leveraging Remote Sensing (RS) and GIS technologies.
- f) Climate Change Impact Assessment – Evaluating effects on surface and subsurface hydrology.

- 3. Profile Summary:** I am a research professional with 3 years + of experience at ICAR-IIWM, Bhubaneswar. I have conducted research on hydrological and groundwater modelling, land use/land cover (LULC) analysis, drought and flood risk assessment, soil erosion modelling, and Energy-Food-Water (EFW) nexus studies across India. I specialise in designing GIS workflows, generating policy-relevant visualisations, and translating complex data into actionable insights for sustainable environmental planning. My work contributes directly to UN Sustainable Development Goals, including SDG 6 (Clean Water and Sanitation), SDG 13 (Climate Action), SDG 15 (Life on Land), and SDG 2 (Zero Hunger), by promoting sustainable water management, climate resilience, and efficient irrigation practices.

4. Technical Skills

- a) GIS & Remote Sensing: ArcGIS Pro, QGIS, Google Earth Engine.
- b) Programming & Analysis: R Studio
- c) Spatial Analysis & Modelling: Watershed hydrology, drought/flood modelling, LULC analysis, soil erosion modelling (RUSLE), groundwater quality (Entropy, TOPSIS)
- d) Visualisation: Map design, dashboards, interactive data storytelling
- e) Tools: GMS, WMS.
- f) Other Skills: Field data collection, analysis of data, research report writing, and multidisciplinary team collaboration.

Curriculum Vitae

5. Academic Qualifications

Year of passing	Degree	Institution	Specialization	%/CGPA
2025	Ph.D.	CUTM	Civil Engineering (Water Resources)	-
2018	M.Tech	NIT Rourkela	Civil Engineering (Water Resources)	87.30%
2012	B.Tech	BPUT	Civil Engineering	68.40%
2008	12th	CHSE	+2 Science	55.5%
2006	HSC	BSE	Matriculation	68%

6. Professional Training

Organization	Duration	Training Area
ACWADAM, Pune	06/01/2014 – 21/01/2014	Planning, Development and Management of Groundwater with Reference to Watershed Management Programs
ISRO (NIT Rourkela)	21/08/2017 – 15/09/2017	Remote Sensing and Digital Image Analysis
GCE Keonjhar	05-02-2019	Technical Writing and Consultancy Advice
VSSUT, Burla	12/03/2019 – 16/03/2019	Water Urbanism
ICAR-IIWM	13/01/2024 – 16/01/2024	Groundwater Modelling System (GMS), Watershed Management System (WMS)
ICAR-IIWM	21-02-2024 – 24-02-2024	Arc GIS Pro

7. Employment Records

Address of Employer/Institution.	Service Period From	Service Period To	Designation	Work Description	Total Period of Employment
F.E.S., DHENKANAL (MGNREGA)	01/11/2012	31/08/2013	Technical Consultant	Works on Natural Resources Development	10 MONTHS
F.E.S. (OTELP+), KEONJHAR	01/09/2013	30/05/2014	Technical Expert	Works on Land And Water Conservation	9 Months
UTCL, BALASORE	31/05/2014	31/05/2015	M.L.E.	Field Work	12 Months
GCE, KEONJHAR	14/08/2018	22/01/2019	Asst. Professor	Teaching	5 Months
GCE, KEONJHAR	24/01/2019	13/06/2019	Asst. Professor	Teaching	5 Months
GCE, KEONJHAR	18/01/2020	15/06/2020	Asst. Professor	Teaching	5 Months
CASA INFRATECH, Bhubaneswar	30/06/2019	15/07/2021	Civil Engineer	Civil Engineer	24 Months

Curriculum Vitae

ICAR IIWM, Bhubaneswar	15/07/2022	Continue	Research Scientist	GIS And Field Visit, And Experimental Work	3year
VANIK Coaching Centre, Bhubaneswar	June2017	Continue	Educator	Guest Faculty	8 Year

8. Research papers in journals:

Publications (SCI/Scopus)

Dandapat, A.K., Panda, P.K., & Sankalp, S. (2024). Quantifying Soil Erosion in Drought-Impacted Central Odisha, India, through Geospatial Mapping with RUSLE. Applied and Environmental Soil Science (Wiley).

Dandapat, A.K., Panda, P.K., Sankalp, S., & Jothimani, M. (2024). Integrated assessment of groundwater quality for sustainable irrigation in drought-prone central districts of Odisha, India. Discover Sustainability (Springer).

Sethi, R.R., Dandapat, A.K., Sankalp, S., & Jena, S.K. (2023). Innovative trend analysis for evaluation of groundwater storage in Baitarani River Basin. Environmental Earth Sciences (Springer).

Sethi, R.R., Dandapat, A.K., Sankalp, S., & Panda, D.K. (2024). Analysing time series variations in groundwater levels within the Baitarani River Basin, India. Environmental Earth Sciences (Springer).

Sethi, R.R., Dandapat, A.K., A.R.H., & Jena, S.K. (2025). Impact of Groundwater Pumping on CO₂ Emissions and Water Productivity in Western Gujarat Agriculture. Water Conservation Science and Engineering (Springer).

Dandapat, A.K., Panda, P.K., & Sankalp, S. (2025). Evaluating drought risks and meteorological hazards in the Mahanadi basin: impact of precipitation deficits and climate change on the downstream regions of Hirakud. Discover Applied Sciences (Springer).

Dandapat, A.K., Sahoo, S.N. (2021). Evaluation of loss models and the effect of LU/LC changes on surface runoff in the Subarnarekha river basin. ISH Journal of Hydraulic Engineering (Taylor & Francis).

Sethi, R.R., Dandapat, A.K., Jena, S.K., & Sankalp, S. (2025). Integrating RUSLE Model with Geographic Information System for Prediction of Soil Erosion in Baitarani River Basin of Odisha, India. Journal of Soil and Water Conservation.

Sethi, R.R., Dandapat, A.K., Sarab, A.R., & Sarangi, A. (2025). Exploring Hydrological and Environmental Transformations by Tank Structures in Odisha Using Sentinel-2 Data. Research Square (Preprint).

Sethi, R.R., Dandapat, A.K., ... (2024). Flood susceptibility mapping using the analytic hierarchy process and GIS in the Baitarani Basin of Odisha, India. Journal of Agricultural Engineering (ISAE).

Conference Proceedings

Dandapat, A. K., Sahoo, S. N. (2017). Evaluation of different loss models for runoff estimation. NIT Rourkela Repository.

Jena, R. K., & Dandapat, A. K. (2019). Estimation of morphometric study on the Shetrunji River using remote sensing and GIS. ISH-Hydro 2019 International Conference.

Dandapat, A. K., Panda, P. K., & Others. (2021). Evaluation of loss models and the effect of LULC changes on surface runoff in Subarnarekha River Basin. GIS Day Celebration 2021, Centurion University, Odisha.

Dandapat, A. K., Panda, P. K., & Sankalp, S. (2022). Determination of R Factor in Revised

Curriculum Vitae

Universal Soil Loss Equation (RUSLE) using R Studio. GIS Day Celebration 2022, Centurion University, Odisha.

Dandapat, A. K., Panda, P. K., & Sankalp, S. (2023). A topographical investigation of twenty-nine geomorphological parameters over the Kuladara watershed, Odisha. AIP Conference Proceedings, 2763(1).

Dandapat, A. K., Panda, P. K., & Sankalp, S. (2023). Analysis of irrigation water quality for agriculture through the EWQI technique. International GIS Day 2023, Centurion University, Odisha.

Dandapat, A. K., & Panda, P. K. (2024). Empowering a sustainable future with geospatial technology. International Seminar on GIS Day 2024, Centurion University, Odisha.

9. Achievements & Awards

- a) GATE Qualified: 2019, 2021, 2023, 2024, 2025
- b) Recognised Contributions: Significant work in GIS-based climate-smart agriculture and sustainable water management projects
- c) Best Poster Awards:
 - (i) ESRI Conference 2024, CUTM, Odisha
 - (ii) System of Crop Intensification for Climate-Smart, Water- and Nutrient-Efficient Agriculture, ICAR-IIWM, Bhubaneswar

10. Name of the references:

Sl. No	Name & Designation	Organization	Address / Location	Email	Mobile No.
a.	Mr Alok Patel, Dept. of Civil Engineering	Government College of Engineering (GCE), Keonjhar	Keonjhar – 758002, Odisha	alokpatel_fce@cekr.ac.in	7008679287
b.	Prof. Sanat Nalini Sahoo, Dept. of Civil Engineering	National Institute of Technology (NIT), Rourkela	Rourkela – 769008, India	sahoosanat@nitrrkl.ac.in	9437572808
c.	Mr Pradeep Kumar Maharana, Team Leader	Foundation for Ecological Security (FES)	Keonjhar, Odisha	pradeepkmaharana@gmail.com	8895643491
d.	Mr Naveen Kumar Nayak, TSE	Ultratech Cement	Balasore, Odisha	naveen20.1984@gmail.com	9937743155
e.	Mr Sumit Pattanayak, Director	CASA Infratech Pvt. Ltd.	Bhubaneswar, Odisha	sumitpattanayk91@gmail.com	9437467746

11. Declaration:

I declare that the foregoing information is correct to the best of my knowledge and belief, and nothing has been concealed/distorted. If at any time it is found to be incorrect/false, my CV is liable to be rejected at any point in time.

Date: 02/10/2025

Place: Bhubaneswar, Odisha

Asit Kumar Dandapat

Signature of the candidate